

Zain Dana Harper

Systems Engineer

Seattle, Washington | Open to remote, hybrid, onsite, contract, and project work zaindharper@gmail.com | <https://harperz9.github.io> | <https://github.com/HarperZ9>

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Summary

Systems engineer and cross-domain builder whose public work spans AI-agent infrastructure, compilers and language tooling, real-time graphics and color science, verification systems, technical documentation, and operations. Builds inspectable tools with explicit boundaries, tests, documentation, provenance, and reproducible release paths. Independent engineering practice since 2023, grounded in earlier technical support, freelance technical writing, compliance documentation, customer-facing operations, and field leadership.

Selected engineering work

Project Telos and model-era engineering tools | Python, TypeScript, Node.js

- Built a public ecosystem for repository intelligence, evidence intake, multi-agent orchestration, evaluation, replayable records, and claim verification.
- Shipped focused tools including Index 2.9.0 for repository maps and context packs, Gather 1.6.1 for provenance-aware research intake, Forum 1.13.0 for model-agnostic orchestration, and Crucible 1.2.0 for worker/verifier evaluation workflows.
- Developed public source packages for model-endpoint coding-agent routing (Relay), toolchain auto-wiring (Plexus), and provenance-carrying agent memory (Mneme).
- Designed CLI, MCP, web, documentation, testing, release, and evidence surfaces so another operator can inspect how a result was produced.

BuildLang compiler and language tooling | Rust, C, HLSL, VS Code

- Built a public lexer-to-code-generation compiler with typed effects, ownership and lifetime analysis, a primary C backend, shader outputs, and editor support.
- Maintained public package and extension surfaces while keeping experimental backends and self-hosting work clearly separated from shipped capability.

Real-time graphics, color, and display tooling | C++, HLSL, Python

- Released a Skyrim real-time graphics project whose current public career materials report more than 900,000 downloads.
- Built D3D11/HLSL post-processing and rendering systems involving tone mapping, TAA, SSR, SSGI, GTAO, volumetrics, shader pipelines, and read-only shared-memory IPC.

Open-source contributions

Upstream pull requests to public projects | Python, TypeScript, Rust

- 22 code, test, and documentation pull requests merged into 19 public repositories maintained by others, including Datasette, tomlkit, pydantic-ai, DeepEval, pydash, and grimp.
- Merged defect work covers HTTP status handling for malformed composite-key row URLs, a TOML round-trip bug where an array of tables swallowed the next sibling key, a metric crash on unhashable tool outputs, missing module and layer validation errors, and lint failures blocking a red main branch.
- Merged test work adds seeded determinism coverage, mid-task recovery fixtures, edge-case fixtures, and evaluation benchmark suites to agent and orchestration projects.
- 14 further pull requests are open and awaiting maintainer review across 13 repositories, including the Model Context Protocol Inspector and Python SDK, Poetry, Drizzle ORM, and LLM. Open is not accepted.

Experience

Independent Systems Engineer / Founder-Builder | Project Telos and HarperZ9 Seattle / remote | 2023-present

- Own architecture, implementation, integration, documentation, testing, packaging, public demos, and release evidence across a multi-repository systems portfolio.
- Build Python CLIs and libraries, Rust compiler tooling, TypeScript and Node workflows, static and interactive web surfaces, C++23 kernels, HLSL graphics systems, and cross-tool adapters.
- Coordinate agentic development workflows while preserving first-party review, verifier separation, testable contracts, and public claim discipline.

Freelance Technical Writer / Documentation and Product Operations Remote | 2017-present

- Produce API and implementation guides, security and compliance documentation, proposals, release notes, support material, and structured technical communication.
- Work with NIST 800-171, CMMC readiness, SOC 2, ISO 27001, DFARS, incident response, and audit-support concepts in a technical-writing capacity.
- Translate implementation detail into material usable by engineers, operators, customers, and nontechnical stakeholders.

Operations and Commercial Arboriculture Lead | Family business Seattle area | 2015-present

- Lead client intake, estimates, site assessment, scheduling, safety judgment, crew and vendor coordination, written proposals, and customer communication.
- Apply practical risk assessment, prioritization, and follow-through in a field environment where plans must survive contact with the job.

Technical Networking Support, Xbox Division | Microsoft Redmond, Washington | 2013-2014

- Diagnosed TCP/IP, DNS, NAT, firewall, router, and account-adjacent console networking issues across phone and chat support.
- Documented repeatable resolutions and translated technical fixes into clear user guidance.

Technical strengths

Languages: Python, Rust, C++, TypeScript/JavaScript, Lua, HLSL, C#, PowerShell, Bash **Systems:** compilers, lexing/parsing, AST and IR design, type checking, code generation, D3D11, shader pipelines, shared-memory IPC, CMake, vcpkg **AI and evaluation:** model routing, tool-use loops, MCP surfaces, multi-agent orchestration, task decomposition, worker/verifier separation, provenance, replayable ledgers, evidence intake **Graphics and color:** real-time post-processing, HDR and tone mapping, ICC and 3D-LUT workflows, perceptual color

difference, DDC/CI, display characterization **Delivery:** Git and GitHub, pytest, Node test tooling, GitHub Actions, release notes, developer documentation, implementation guides, troubleshooting, Linux

Public proof

Portfolio: <https://harperz9.github.io/portfolio.html> Research: <https://harperz9.github.io/research.html>
Repositories and releases: <https://github.com/HarperZ9>

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